United States Patent	12397839
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Winden; Steve

Hand truck automated load unload

Abstract

An apparatus for interconnecting a hand truck or dolly to a vehicle receiver or hitch while hauling a heavy load. The apparatus includes a base, an extendable lever arm and a separable intermediary glide. The base couples with the hand truck, the lever arm is extendable from the base, and the intermediary glide interconnects the lever arm to a receiver attached to a vehicle. In accordance with certain aspects of embodiments of the invention the hand truck and glide may be automated to automatically stabilize the hand truck, lift it into alignment with the receiver, and then lock the lever arm into place relative to the receiver and vehicle. Further, the system in accordance with aspects of the invention may automatically unload the hand truck from engagement with the vehicle.

Inventors: **Winden; Steve (Mesa, AZ)**

Applicant: **Winden; Steve** (Mesa, AZ)

Family ID: **1000008781777**

Appl. No.: **17/669284**

Filed: **February 10, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220161836 A1	May. 26, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 16727981 20191227 US 11305802 child-doc US 17669284
us-provisional-application US 62848021 20190515

Publication Classification

Int. Cl.: **B62B5/00** (20060101); **B60R9/06** (20060101)

U.S. Cl.:

CPC **B62B5/0003** (20130101); **B60R9/06** (20130101);

Field of Classification Search

CPC: B62B (5/0003); B62B (1/12); B62B (1/10); B60D (1/065); B60D (1/52); B60D (2001/005); B60R (9/06); B60R (2011/004); B60R (2011/0078); B60R (11/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5181822	12/1992	Allsop et al.	N/A	N/A
5845831	12/1997	Nusbaum et al.	N/A	N/A
6155588	12/1999	Maxey	N/A	N/A
6237823	12/2000	Stewart et al.	N/A	N/A
6314891	12/2000	Larson	N/A	N/A
7641235	12/2009	Anduss	N/A	N/A
8123238	12/2011	Burgess	N/A	N/A
8210559	12/2011	Russell	N/A	N/A
8480149	12/2012	Durand	N/A	N/A
8505951	12/2012	Bohse	N/A	N/A
9096160	12/2014	Le Anna	N/A	N/A
9216698	12/2014	Rhodes	N/A	N/A
10059276	12/2017	Phillips	N/A	N/A
10112523	12/2017	McConn et al.	N/A	N/A
10189419	12/2018	Billard	N/A	N/A
10315474	12/2018	Reynolds	N/A	N/A
11305802	12/2021	Winden	N/A	B60D 1/52
2004/0173654	12/2003	McAlister	N/A	N/A
2008/0101899	12/2007	Slonecker	N/A	N/A
2010/0066069	12/2009	Bradshaw	N/A	N/A
2019/0168555	12/2018	Axelson, Jr.	N/A	N/A

Primary Examiner: Neacsu; Valentin

Assistant Examiner: Hymel; Abigail R

Attorney, Agent or Firm: Dietz Law Office LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the filing benefit and priority of U.S. Provisional Patent Application Ser. No. 62/848,021 filed on May 15, 2019 and U.S. Non-Provisional patent application Ser. No. 16/727,981 filed on Dec. 27, 2019 the contents of which are incorporated herein by reference in their entireties.

FEDERAL SPONSORSHIP

(1) Not Applicable

JOINT RESEARCH AGREEMENT

(2) Not Applicable

TECHNICAL FIELD

(3) The present invention relates to an interconnection between a vehicle and a connecting hitch. More particularly, the invention relates to a hitch or receiver and receiver interconnection that is utilized to join a hand truck or dolly to a vehicle. The receiver assembly in accordance with the present invention allows a user to wheel a loaded dolly to a vehicle and secure the dolly to the vehicle receiver for transport, without requiring ramps or a linear vertical lifting of the dolly to a height of the vehicle's hitch receiver. In accordance with certain aspects of embodiments of the invention the receiver and hand truck may include an automated system to stabilize the hand truck, lift it into alignment with the receiver, and then lock it into place relative to the receiver and vehicle. Further, the system in accordance with aspects of the invention may automatically remove the hand truck from locked engagement with the vehicle.

BACKGROUND

(4) Over the years various mechanism have been contrived to ease the burden of moving a heavy load. By way of example, a person may desire to load a piece of heavy goods into a vehicle's rearward compartment, box or bed. To move the heavy goods a person may utilize for example, ramps and a hand truck, cranes, lift gates, multiple people to lift the goods safely to the vehicle, or other hoisting mechanisms. Use of these mechanisms may require additional space within a vehicle to allow a person to load and unload the heavy goods. Further, use of these mechanisms often requires more than one person to load or unload the heavy goods. External cargo racks have been attached to a vehicle receiver, however, when loading the cargo rack vertical lifting of the load is required to load the cargo rack at the height of the vehicle receiver.

(5) Alternatively, a ramp may be required to move a heavy load from the ground to the height of the cargo rack engaged to the vehicle receiver. The present invention provides an apparatus that one person may use without ramps to secure and remove heavy equipment from an exterior of a vehicle while utilizing a hitch receiver attached to the vehicle. Further, the invention allows one user to couple and uncouple a hand truck to a vehicle hitch receiver without the need for ramps, lift gates, or multiple persons. Additionally, the invention includes a lock mechanism that ensures that portions of the invention do not unintentionally pull out of the vehicle while unloading the hand truck.

SUMMARY

(6) Embodiments of the invention include a device to allow a single person to be able to load/unload a hand-truck or dolly to/from a receiver or hitch while reducing the required lifting force. Embodiments of the invention further automate the loading and unloading the hand truck to/from the vehicle's receiver. The device remains engaged to the hand truck when the hand truck is coupled to the vehicle receiver. The apparatus attaches to a hand-truck and allows for a “shaft” to reach up to vehicle receiver while the hand truck is at ground level. The vehicle receiver may be incorporated into a trailer hitch receiver or may be attached to a vehicle independent of the vehicle's trailer hitch. The shaft attaches to a “glide” inserted into the receiver allowing the load secured on hand truck to be raised off the ground, then slid into the receiver for secure attachment to the vehicle. The glide may be coupled to a drive mechanism to further assist the user in loading and unloading the hand truck.

(7) According to aspects of the invention the apparatus is well suited for connecting a hand truck or dolly to a receiver attached to a vehicle. The apparatus includes a base, a lever arm and a glide. The

base has a hand truck receiving portion to which a hand truck may be attached. The lever arm extends from the base. A free end portion of the lever arm extends outward from the base and terminates in a claw. The free end portion is adapted to be received within the receiver. The glide is separable from the lever arm and receiver, wherein the glide is adapted to be received within the receiver. The glide has a pivot member that is adapted to couple with the claw of the lever arm.
(8) Those skilled in the art will appreciate that the lever arm may instead include the pivot member and the claw incorporated into the glide. A vehicle's trailer hitch receiver may be modified so that the receiver is long enough to contain the glide and a portion of the lever arm when the dolly is connected to the vehicle.
(9) Embodiments according to aspects of the invention may additionally include an end portion of the lever arm that is slidingly extendable from the base. Also, a lever lock may be coupled to the base in a manner having two positions: a first locked position that engages the lever arm and a second extending position that disengages the lever arm from the lock. The base may further include a storage slot formed in the base that is well suited and adapted for storing the glide within the base. The glide may further include a button ball release that compresses into the glide and actuates outward from the glide when a compressing force is not present. The button ball release engages a hole formed in the hitch receiver for the hitch pin. When compressed, the glide may be slid further into the hitch receiver. Low friction wear plates may be attached to sides or top and bottom of the glide and lever arm to facilitate smooth sliding of the lever arm and glide within the hitch receiver. The lever arm may further include an aperture extending through a sidewall of the lever arm, wherein the aperture adaptable to receive a hitch pin. The hitch pin may be used to lock base and lever arm to the hitch receiver of the vehicle. In this manner, a hand truck may be locked in place to the back end of a vehicle.
(10) A further embodiment according to aspects of the invention includes a base, lever arm and glide. The base is adapted to couple with a hand truck or dolly. The lever arm is extendable from the base, wherein the lever arm has an end portion slidingly engaged with the base and a free end portion extending outward from the base. The free end portion includes a claw or pivot wherein the free end portion is adapted to be received within a trailer hitch receiver. The glide is separate from the lever arm and trailer hitch receiver, wherein the glide is adapted to be received within the trailer hitch receiver and has a pivot member or claw that is adapted to couple with the claw or pivot of the lever arm.
(11) Embodiments according to aspects of this invention may further include a lever lock coupled to the base. The lock has a first locked position that engages the lever arm and has a second extending position that disengages the lever arm from the lock. When the lock is in the extending position the lever arm may slide outward from the base. A storage slot may be formed in the base and dimensioned to receive and store the glide within the base. The glide may further include a compressible button ball release that compresses inward into a hollow portion the glide. The button ball release is sized to engage within a hitch pin hole formed in the vehicle receiver. When engaged, the button ball retains the glide in a fixed position within the vehicle receiver. A user may compress the button ball release and slide the glide further into the vehicle receiver. The lever arm includes an aperture extending through a sidewall of the lever arm, wherein the aperture adaptable to receive a hitch pin. The claw of the lever arm may engage the pivot pin of the glide. The base may be pushed such that the lever arm presses against the glide and slides into the vehicle's receiver until the aperture of the lever arm aligns with the hitch pin hole of the receiver.
(12) A still further embodiment according to aspects of the invention includes a base, lever arm and glide that act together to connect a hand truck to the trailer hitch receiver of a vehicle. The base includes a hand truck receiving portion that couples or is fixed to a hand truck. The lever arm slides within a channel formed in the base and may extend between a stowed position and a lift position. The lever arm has a rearward end portion that slides within the base. On the opposite end of the lever arm is a free end portion that extends outward from the base and includes a claw or pivot

formed on the end portion. The free end portion and claw or pivot are adapted to be received within the trailer hitch receiver. The glide is separable from both the lever arm and trailer hitch receiver. The glide is adapted to be received within the trailer hitch receiver. The glide has a mating pivot member or claw adapted to couple with the respective claw or pivot of the lever arm.

(13) The glide also includes a compressible button ball release that engages the trailer hitch receiver to lock the glide in the receiver during loading. Additionally, a lever lock is coupled to the base. The lock has a first locked position that engages the lever arm to restrict movement of the lever arm within the base. The lock has a second extending position that disengages the lever arm from the lock to allow the sliding and extension of the lever arm from the base. Low friction wear plates may be attached to sides of the glide and lever arm to reduce the required force to push the glide and lever arm into the receiver.

(14) In accordance with further aspects of the invention the base may be solidly attached (welded) to a hand-truck or may be fastened or otherwise coupled to the tongue portion of a hand truck. The lever arm or shaft protrudes and retracts into base to allow for increased vertical reach when mounting the hand-truck to the vehicle hitch. The glide has ball catches that may be used to temporarily hold the glide in place when attempting to hook up the lever arm or shaft to the glide. The glide is designed to reduce drag when the weight of the load on the hand-truck transfers to glide. Also, the lever arm or shaft is designed to reduce drag while sliding in and out of the receiver under load, while maintaining tight clearances for exacting operation. The lever arm or shaft is designed to lock into an extended position while the hand truck is tilted back and engaged to the glide. The lock is spring loaded to allow the lever arm or shaft to retract into body and automatically stop when in mounted or travel ready position.

(15) A compartment or slot is formed in the base allowing the glide to be stored in the body or base in a secure location during storage. The glide slides out of receiver when unloading, stopping at hitch pin hole when the compressible ball catches engage the hitch pin holes. Once the glide is locked the hand cart may be tilted until the wheels reach the ground. Those skilled in the art will appreciate that the apparatus of the invention is not limited to any specific size dolly or hand-cart and the apparatus can be scaled to accommodate different load sizes and weights. The lever arm is fixed to the vehicle receiver with a hitch pin and is not limited to size or hole setback distance. The base and hand cart allow for multiple methods of securing a load to a vehicle.

(16) In accordance with aspects of the invention, an embodiment of the invention includes an apparatus for autonomously connecting and disconnecting a hand truck to a vehicle with minimal user interaction. The apparatus includes a hand truck, a lever arm, a receiver, a glide, and a drive mechanism. The lever arm is coupled to a base of the hand truck. A free end portion of the lever arm extends outward from the hand truck. The receiver is adapted for coupling to a vehicle. By way of example, the receiver may be welded to the vehicle or may couple with a receiver already fixed to the vehicle. The glide is contained within the receiver and slides within the receiver between a storage position, a load position (the glide partially extends from the receiver), and a retracted position (the glide is fully contained within the receiver).

(17) The drive mechanism is fixed or coupled to the glide and actuates to linearly displace the glide within the receiver between the loading position and the retracted position. A free end of the glide and lever arm link together allowing the glide to pull the lever arm into the receiver when the drive mechanism actuates the glide inward. Similarly, when the drive mechanism actuates the glide outward, the glide pushes against the lever arm and forces the lever arm out of the receiver. Without limitation intended, the linkage between the free end of the lever arm and the glide may include a pivot and claw arrangement. Either the glide or lever arm includes a claw and then the other of the glide and lever arm has a pivot member that is adaptable to link or couple with the claw.

(18) In accordance with further aspects of the invention, two spaced apart winches are attached to either the hand truck or the vehicle. Hooks at the end of each winch line are connected to the

opposing vehicle or hand truck. When the lever arm is linked to the glide the winches spool in the winch line which cause the hand truck to rotate upwards. A controller is electrically linked with the winches and drive mechanism, wherein the controller provides a signal to control coordinated activation of the winches and drive mechanism. Further, sensors are positioned on the hand truck and receiver, which provide signal outputs to the controller. The signal output is processed within the controller to determine and correlate a three-dimensional orientation of the hand truck and receiver. By way of example and without limitation intended, the device orientation may be determined using device orientation sensors such as micromechanical gyroscopes or accelerometers and geomagnetic field sensors. With output from the device orientation sensors the controller controls activation of the winches to stabilize the hand truck and orient the hand truck to align the length axis of the lever arm with the longitudinal axis of the receiver. In certain embodiments the drive mechanism includes a linear actuator that has a lead screw, lead nut, and stepper motor of known suitable construction.
(19) The accompanying drawings, which are incorporated in and constitute a portion of this specification, illustrate embodiments of the invention and, together with the detailed description, serve to further explain the invention. The embodiments illustrated herein are presently preferred; however, it should be understood, that the invention is not limited to the precise arrangements and instrumentalities shown. For a fuller understanding of the nature and advantages of the invention, reference should be made to the detailed description in conjunction with the accompanying drawings.

Description

DESCRIPTION OF THE DRAWINGS
(1) In the various figures, which are not necessarily drawn to scale, like numerals throughout the figures identify substantially similar components.
(2) FIG. **1** is a top front perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;
(3) FIG. **2** is a left side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;
(4) FIG. **3** is a right side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;
(5) FIG. **4** is a left back perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled and having a portion of a hand truck fixed to the base;
(6) FIG. **5** is a bottom front perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;
(7) FIG. **6** is a bottom left side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;
(8) FIG. **7** is a bottom right side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;
(9) FIG. **8** is a partial section bottom perspective view of an apparatus for connecting a hand truck

to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base and glide aligned but uncoupled;

(10) FIG. **9** is a bottom perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base tilted and coupled to the glide and showing the glide aligned but uncoupled to the receiver;

(11) FIG. **10** is a partial sectional bottom perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle of the type shown in FIG. **9**, showing the base tilted and coupled to the glide and showing the glide aligned but uncoupled to the receiver;

(12) FIG. **11** is a partial exploded top plan view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

(13) FIG. **12** is a partial exploded side perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

(14) FIG. **13** is a partial exploded front bottom perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

(15) FIG. **14** is a partial exploded side view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

(16) FIG. **15** is a partial exploded bottom side perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

(17) FIG. **16** is a partial exploded top side perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

(18) FIG. **17** is a perspective view of a hand-truck and lift device in accordance with an embodiment of the present invention shown in a pre load position;

(19) FIG. **18** is a perspective view of the hand-truck and loading device of the type shown in FIG. **17** with the receiver glide inserted into the receiver;

(20) FIG. **19** is a rear perspective view of the hand truck and loading device of the type shown in FIG. **18**;

(21) FIG. **20** is a perspective view of the hand truck of the type shown in FIG. **18** and illustrated as tilting back at an approximately 40° angle in preparation to attach the lever arm or shaft to the glide or receiver glide;

(22) FIG. **21** is a side perspective view of the hand truck base of the type illustrated in FIG. **20** and shown tilted back and attached or coupled to the glide device;

(23) FIG. **22** is a rear perspective view of the hand truck of the type illustrated in FIG. **21**;

(24) FIG. **23** is a perspective view of the hand truck, base and lever arm in accordance with the present invention and shown lifted up to the same height or level as the receiver;

(25) FIG. **24** is a bottom perspective view of the invention of the type illustrated in FIG. **23**;

(26) FIG. **25** is a perspective view of the invention of the type illustrated in FIG. **23** and further showing the lever arm retracted into the body or base of loading device of the present invention;

(27) FIG. **26** is a bottom perspective view of the device illustrated in FIG. **25** illustrating the hand truck lifted up and the lever arm or shaft retracting into the body or base of loading device;

(28) FIG. **27** is a perspective view of the apparatus in accordance with the present invention showing the lever arm or shaft fully inserted into a vehicle receiver and ready to secure with hitch pins and straps;

(29) FIG. **28** is a bottom perspective view of the invention of the type illustrated in FIG. **27**;

(30) FIG. **29** is a side perspective view illustrating a glide sticking out of receiver and having the

lever arm or shaft attached to guide;
(31) FIG. **30** is a side perspective view of an apparatus in accordance with aspects of the invention showing the lever arm attached to glide and ready for a load to be raised and slid into vehicle receiver;
(32) FIG. **31** is a side perspective view of an apparatus in accordance with aspects of the invention showing the hand truck coupled to the vehicle receiver (in a loaded position) and secured at a back of vehicle receiver and ready for transport;
(33) FIG. **32** is a back perspective view of a hand truck in accordance with aspects of the invention showing winch lines extending from winches coupled to the hand truck;
(34) FIG. **33** is a side perspective of a drop down style auto lift/load receiver of the present invention;
(35) FIG. **34** is a perspective view showing an auto lift/load receiver of the present invention shown fixed to a rear portion of a vehicle frame;
(36) FIG. **35** is a perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a preload orientation;
(37) FIG. **36** is a side perspective view of a hand truck in accordance with aspects of the invention;
(38) FIG. **37** is a side perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation;
(39) FIG. **38** is a perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation;
(40) FIG. **39** is a side perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated orientation and having the lever arm axis aligned with the longitudinal axis of the receiver;
(41) FIG. **40** is a perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated orientation and having the lever arm axis aligned with the longitudinal axis of the receiver;
(42) FIG. **41** is a rear perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated orientation and having the lever arm axis aligned with the longitudinal axis of the receiver;
(43) FIG. **42** is a partial section perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in a fully stowed or travel orientation;
(44) FIG. **43** is a side perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation and having a block and tackle configuration incorporated with the winches;
(45) FIG. **44** is a partial section perspective view of a hand truck in accordance with aspects of the invention and shown having a block and tackle configuration incorporated with the winches;
(46) FIG. **45** is a perspective view of a winch embodiment in accordance with aspects of the invention;
(47) FIG. **46** is a perspective view of a controller embodiment in accordance with aspects of the invention;
(48) FIG. **47** is a right side perspective view of a drop down style auto lift/load receiver of the present invention;
(49) FIG. **48** is a left side perspective view of a drop down style auto lift/load receiver of the present invention;
(50) FIG. **49** is a partial section perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a preload orientation;
(51) FIG. **50** is a partial section perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation;
(52) FIG. **51** is a partial section perspective view of a lever arm aligned with an auto lift/load

receiver in accordance with aspects of the invention and shown in an elevated and aligned orientation;
(53) FIG. **52** is partial sectional perspective view of a drop down style auto lift/load receiver of the present invention;
(54) FIG. **53** is a partial sectional perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a preload orientation;
(55) FIG. **54** is a partial sectional perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated and aligned orientation;
(56) FIG. **55** is a partial sectional perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a fully stowed or travel orientation;
(57) FIG. **56** is a perspective view of a portion of the lever arm and glide in accordance with aspects of the invention and shown linked together; and
(58) FIG. **57** is a partial section perspective view of a portion of the lever arm and glide in accordance with aspects of the invention and shown linked together.

DETAILED DESCRIPTION

(59) The following description provides detail of various embodiments of the invention, one or more examples of which are set forth below. Each of these embodiments are provided by way of explanation of the invention, and not intended to be a limitation of the invention. Further, those skilled in the art will appreciate that various modifications and variations may be made in the present invention without departing from the scope or spirit of the invention. By way of example, those skilled in the art will recognize that features illustrated or described as part of one embodiment, may be used in another embodiment to yield a still further embodiment. Thus, it is intended that the present invention also cover such modifications and variations that come within the scope of the appended claims and their equivalents.
(60) The hand truck vehicle mounting assembly **10** of the present invention is particularly well suited for externally coupling a hand truck or dolly to a receiver mounted to a vehicle. In some embodiments a vehicle's trailer hitch receiver or a trailer hitch receiver having an extended receiver is suitable for use with the invention. The present invention is further particularly well suited to allow a single user to couple a heavily loaded dolly to the vehicle hitch without requiring simple vertical lifting of the dolly to the height of the receiver. By way of example, a loaded dolly is tilted backward and then positioned so that a lever arm engages with a glide that is engaged with the vehicle hitch receiver. As the user rotates the dolly upward towards the vehicle, the glide slides into the receiver and the lever arm also slides into the receiver. The lever arm and glide of the present invention reduces the amount of lifting force required to lift the dolly to the height of the vehicle receiver.
(61) Turning attention now to the Figures, embodiments according to aspects of the invention will now be described. Referring first to FIGS. **1-7**, the hand truck vehicle mounting assembly **10** generally includes a base **40**, lever arm **100**, and glide **150** that may couple to a vehicle hitch receiver **200**. In an embodiment of the invention the base has a hollow interior **42** with dividing walls **44** extending between the sides of the base to provide stability and support to the base. The top surface **48** of the base is particularly well suited for supporting a load. The bottom of the base includes an attachment flange **46** extending inward from the sidewalls. Base plate **52** may be coupled to the flange **46** with a weld, fasteners, or other known suitable coupling. The base plate may be further engaged and coupled to the tongue of the dolly or may form a portion of the tongue of the dolly. Further, a back side of the base **40** may include a backing plate **50** or, alternatively, a portion of the dolly may form or establish the back side of the base **40**. The front side of the base includes an opening into a storage compartment **56** for the glide **150**. Also internal to the base is a channel or tube **54** into which the lever arm **100** slides between a stowed position and extended

position. Front extension stop **136** is formed on or coupled to lever arm **100**. A blocker member **134** is attached to the front of the base at the opening of the channel or tube **54**. The lever **100** is slidable through the blocker member **134** until the stop **136** contacts blocker **134**. A rear extension stop **132** on the lever arm **100** inhibits the lever arm from being pulled completely out of the channel **54**.

(62) The lever arm **100** includes sides **102**, side wear guides **106**, top wear plate **108** and bottom wear plate **110**. Free end **120** of the lever arm tapers to a nose **124**. A claw **122** is formed on the lower portion of the free end **120** of the lever arm **100**. Multiple hitch pin holes **130** extend from one side of the lever arm to the other. The multiple holes **130** allowing a user to orient the lever arm **100** relative to the base **40** and hitch receiver **200** in a variety of orientations. Independent glide **150** includes sidewalls **152**, a hollow interior (not shown), a compression release button **156** and a latch **158** (that is pushed out of the glide via spring **160**). An end of the glide **150** includes a pivot pin **170** to which the claw **122** engages. The glide **150** further includes top wear plate **176** and bottom wear plate **178**. In another suitable embodiment, the wear plates may be eliminated by making the glide **150** from a solid wear resistant core (such as nylon) and having side plates and the pivot pin mounted to the core.

(63) The wear plates **106**, **108**, **110**, **176** and **178** may be constructed from a variety of known materials having wear resistant characteristics when sliding against metal and further having coefficients of friction that are lower than a metal surface. One such known suitable material is a carbon fiber filled nylon. The hitch receiver **200** includes tubular sidewalls **202** and a hitch pin hole **204** extending through both sides of the sidewalls **202**. Throughout many of the Figures a ground or surface level is identified with numeral **16** and the vehicle or addition portions of the vehicle receiver frame are identified with dashed lines and numeral **230**. Additionally, the orientation of the base relative to the ground **16** and height of the receiver **200** relative to the ground **16** are depicted in several of the Figures.

(64) Referring next to FIGS. **8-10**, lock pin **60** is mounted in base **40** with mounts **62**. Lock pin **60** is able to slide back and forth through the mounts **62**. An end of the pin **60** extends through lock pin hole **70** formed in the lever arm channel or tube **54** of the base **40**. The end of the pin **60** engages and disengages within holes **130** extending through lever arm **100**. Stop **66** retains spring **64** which returns the lock to the locked position. Groove **68** extends around the lock pin **60** and may be utilized to temporarily restrict the lock pin in a disengaged position. Lock pin cover or plate **72** is mounted or fastened to the bottom flange **46** of the base **40** to enclose the lock mechanism.

(65) FIGS. **11-16** illustrates a linear alignment of the lever arm channel or tube **54** of the base **40**, the lever arm **100**, the glide **150** and the vehicle hitch **200**. The lever arm **100** is shown completely removed from the base to further illustrate components of the lever arm **100** that are contained within the base **40**. A portion of the lever arm **100** slides within and is contained by the lever arm channel or tube **54**. When the lever arm is installed within the lever arm channel **54**, a stop **132** having a perimeter larger than a perimeter of the hole extending through the blocker **134** restricts the lever arm from being removed completely from the channel. The blocker **134** may be attached to or formed as a part of the base **40**. A front extension stop prevents or restricts the lever arm from sliding into the base further than where the stop **136** is positioned on the lever arm **100**. Although two holes **130** are shown extending through the lever arm between the rear extension stop **132** and the front extension stop **136**, those skilled in the art will appreciate that if desired more holes may be created in the lever arm. The holes **130** between the two stop are utilized in conjunction with the lock pin **60** to lock the lever arm into position having a desired length extending from the base. When two holes **130** are provided between the stops, the rearward hole may be utilized to lock the lever arm **100** in the extended or loading position and the forward hole **130** may be utilized to lock the lever arm in the retracted or travelling position. The hole **130** near the free end **120** of the lever arm **100** may be aligned with the hitch pin hole **204** of the hitch receiver **200**. In this manner a hitch pin may be utilized to lock the lever arm **100** to the receiver **200**.

(66) FIGS. **17-28** illustrates the use of the assembly to mount a load of a hand truck **20** to a vehicle hitch receiver **200**. The hand truck **20** includes a loading handle or grip **22** to assist a user when pushing or pulling on the hand truck to slide the lever arm **100** and glide **150** fore and aft. FIG. **18** illustrates a hand truck **20** coupled to the base **40** together with a glide **150** both of which are positioned near a vehicle hitch **100**. When a user desires to connect the dolly to a vehicle, a user then inserts the glide into the vehicle hitch and pushes the glide **150** into the receiver **200**. As the glide **150** is pushed into the receiver latch **158** actuates inward against the compression force of spring **160**. As the glide is pushed inward into the receiver **200** the latch engages with hitch pin hole **204**. The latch includes an angled end portion so that as the user continues to push the glide inward the latch pin **158** will push against the compression force of spring **160** and recede into the glide. The user continues to push the glide **150** inward until the compression buttons **156** engage within the holes **204** of the hitch receiver **200** (see FIGS. **19** and **20**). Next the user tilts the hand truck **20** backwards (FIG. **21**) and moves the hand truck **20** towards the vehicle hitch **200**. The dolly **20** is tilted enough so that the claw **122** is higher than the pivot pin **170** of the glide **150**. Once the claw **122** is positioned over the pivot pin **170** the dolly is tilted forward until the claw engages the pivot pin (FIGS. **22** and **23**). The dolly **20** is then pivoted or rotated upwards from the ground **16** until the lever arm **100** is in linear alignment with the glide **150** and hitch receiver **200** (FIGS. **24** and **25**). Once in linear alignment the lock pin **60** is disengaged the user may apply a force against the hand truck to cause the lever arm **100** to slide into the base **40** (FIG. **26**), the free end **120** slides into the receiver until the front stop **136** engages the receiver (FIG. **27**) and the glide slides further into the receiver **200**. As the slide **150** slides further into the receiver **200** the buttons **156** compress from the sliding force. The compression button **156** disengages from the hitch hole **204** and the glide slides further into the receiver until the front extension stop **136** of the lever arm **100** engages with the hitch receiver **200** (FIGS. **27** and **28**). A hitch pin may then be inserted through hitch pin hole **204** and lever arm hole **130** to secure the hand truck vehicle mount assembly **10** to the vehicle.

(67) When the user wishes to remove the dolly from the vehicle the hitch pin is removed and the user pulls the dolly away from the vehicle. As the glide is pulled from the receiver the compression buttons **156** remain compressed inward against the sidewalls and compress as they pass through the hitch pin hole **204**. The glide continues to pull from the receiver until latch **158** engages the hitch pin hole **204** so that the glide does not pull completely from the receiver **200**. The lock pin **60** on the base **40** is disengaged and the dolly **20** is further pulled outward and away from the vehicle so that the lever arm **100** extends to its extended position from the base **40**. The dolly **20** and base **40** may then pivot downward until the wheels contact the ground **16**. The dolly is then tilted further to disengage the claw **122** from the pivot pin **170**.

(68) FIGS. **29-31** further illustrates the use of the assembly to mount a load of a hand truck to a vehicle hitch receiver in conjunction with adjustable straps. Although a load on the base **40** and dolly **20** are not illustrated, those skilled in the art can appreciate how a load may be positioned on the base **40** and secured to the dolly **20** with straps **30** and hooks **32**. Alternatively, ratchet straps may be used to assist a user when rotating the dolly and base upwards.

(69) With reference to FIGS. **32-57** additional embodiments according to aspects of the invention will be described. With reference first to FIGS. **32-34** the hand truck and glide include additional features that significantly automated the lifting and loading of the hand truck onto the vehicle. The hand truck **20** includes spaced apart winches **28**, a controller **80** and battery or power supply **98** fixed to the hand truck. The winches include load straps or cable **30** wound on a spool contained within a spool housing **94**. The spool is rotated by a winch motor **96** to thereby extend or retract the cable through a cable outlet **92** formed in the housing **94**. FIG. **45** further illustrates components of the winch **28**.

(70) The winch is powered by battery **98** and operation of the winches may be achieved with controller **80**. The controller **80** includes a controller housing **90** in which a processor and circuitry

are contained. The controller **80** is electrically coupled to the winches **28** and power supply **98**. The controller includes a joystick **88** for user controlled operation of the lifting and loading of the hand truck to the vehicle **228**. The controller also includes power switch **82**, auto load activation switch **84**, and quick stop switch **86**. FIG. **46** further illustrates components of the controller **80**.

(71) The lever arm **100** of the hand truck **20** couples to the auto glide or auto lift/load receiver **300** of the present invention. The lift load receiver **300** generally includes a main receiver housing **306**, linear actuator or motor **314**, glide carriage **330**, and electronic controls **320**. In certain embodiments according to aspects of the invention the lift load receiver **300** may include a drop down receiver **304** which couples to a receiver **232** of a vehicle **228** or may mount directly to the vehicle receiver frame **230** (as illustrated in FIG. **34**). The electronic controls **320** includes a power switch **320**, stop switch **322**, and glide actuator switch **324**. Voltage amplifier **328** is electrically coupled to the linear actuator **314** and electronic controls **320**. FIGS. **47** and **48** further illustrate components of the lift load receiver **300**.

(72) With reference to FIGS. **35-42** the auto load sequence will next be described. The controller **80** on the hand truck **20** and electronic control **320** coupled to the auto lift/load receiver **300** are wirelessly synced to coordinate the lift and load operation. A user may activate the system with a suitable cell phone app or by activating switches on the controllers. Without limitation intended a bluetooth connection between the hand truck and the auto lift/load receiver uniquely communicates between single units avoiding cross talk from other nearby systems. When a user activates the load switch on the hand truck, the auto receiver will automatically load the hand truck to the back of a vehicle. That same single activation process will also unload the device from the back of the vehicle. Those skilled in the art will appreciate that the controllers may include multiple sensors and other electrical components such as proximity switches, piezo sensors, magnetic reed switches, device orientation sensors (for example, micromechanical gyroscopes or accelerometers and geomagnetic field sensors) to perform the various functions herein described. For example, output from sensors are processed to determine and align the 3-axis orientation of the hand truck with the 3-axis orientation of the lift load receiver attached to a vehicle. When the vehicle is on a slope, the controller compensates for both device orientations to align them together.

(73) The user positions the hand truck **20** so that a free end of the lever arm **100** is below the pivot **338** of the receiver **300** (See FIGS. **36** and **36**). The hand truck **20** is then tilted backward and moved forward so that the claw **122** of the lever arm **100** engages the pivot pin **338** of the glide carriage **330** of the receiver **300**. The user spools out enough cable from each winch **28** so that each hook **32** may be engaged to the d-loops or other fixtures **34** fixed to the vehicle **228** (see FIGS. **37** and **38**). The joystick **88** is then used to retract cables **30** so that a slight tension is on the cable. With the power switches **82** and **326** activated, the user may then initiate the load sequence by activating the auto load switch **84** on the controller **80**. The controller **80** activates the winches **28** and retracts the cable in a controlled manner such that the hand truck remains in an upright orientation. As the cables **30** retract, claw **112** rotates about pivot **338** thereby raising the attached hand truck **20** into the air. The controller **100** and winches **28** continues to retract the cables **30** until the longitudinal axis of the lever arm is linearly aligned with the longitudinal axis of the lift load receiver **300** (see FIG. **39-40**). Once the axes are aligned **366** the winches **28** are again activated while the linear actuator **314** is also activated. As will be described in greater detail below, activation of the linear actuator **314** pulls the hand truck **20** into locked engagement position **364** with the vehicle **228** (see FIGS. **41** and **42**). The cable **30** may be further tightened so that the load is tight to the back of the vehicle. To unload the hand truck the controller controls the winches and linear actuator in reverse. Once the cart is removed from the glide, the movable glide retracts into the receiver to a storage position. When user wants to load the device, the user presses a button on glide to extend the glide from the ‘storage’ position **360** to an ‘extended’ position **362**.

(74) FIGS. **43** and **44** illustrate the use of block and tackle **36** and an alternative mounting of the winches **28** and d-loops **34** without departing from the scope of the present invention. FIGS. **49-51**

further illustrates the orientation and alignment of the lever arm **100** relative to the lift/load receiver **300** during a load or unload sequence. During preload, the hand truck is tilted backwards. This causes an angled orientation of the lever arm **100** relative to the lift load receiver **300** (See FIG. **49**). The lever arm includes a claw **122** that engages with the pivot pin **338** coupled to the glide carriage **330**. Guide plates **340** assist a user to align the claw with the pin (See FIG. **50**). Electronic controller communicates with the hand truck controller **80** and receives power from the voltage amplifier **328**. Dust cover **308** restricts dust from entering the interior of the main receiver housing **308**. Once the hand truck is lift or elevated into the load position the axis of the lever arm **80** and receiver **300** are aligned (see FIG. **51**).
(75) Referring next to FIGS. **52-55**, the partial cross section of the receiver **300** cross section of autoload sequence further illustrates the orientation and alignment of the lever arm **100** relative to the lift/load receiver **300** during a load or unload sequence. The receiver **300** includes main receiver housing **306**. Glide carriage **330** slides or rolls within the receiver housing **306** along roller bearings **332**. The glide carriage **330** is actuated within the receiver via linear actuator or stepper motor **314**. The linear actuator is coupled to a lead drive screw **310**. An enclosed end **334** of the glide carriage is coupled to a drive screw lead nut **312**. Without limitation intended, the drive screw extends into a screw tube **318** to thereby allow approximately 6 inches of travel by the glide carriage **330** within the receiver housing **306**. As the drive screw **310** turns within the lead nut **312** the glide is pushed or pulled within the housing. The electronic controller controls actuation of the linear actuator **314**. Stop switch **322** allows a user to stop the actuation as desired. The actuate glide switch **324** may be activated to orient the glide so that the pivot pin **338** at the lever arm end **336** of the glide **330** extends slightly from the receiver housing **306**. An engagement switch **344** is activated when the lever arm claw **122** engages the pivot pin **338**. Sensor **346** further provides feedback to the controller regarding the operation of the lead screw **310**. The glide **330** actuates between a storage position **36** (see FIG. **52**), a load/unload position **362** (see FIGS. **53** and **54**) and a locked and loaded position **364** (see FIG. **55**).
(76) Those skilled in the art will appreciate that the pin and claw arrangement may be reversed such that the free end of the lever arm **100** includes a pivot pin **126** and a free end **336** of the glide carriage **330** includes claw **342** (see FIGS. **56-57**). Additional modifications to the various components of the invention may be altered without departing from the intended scope of the invention.
(77) These and various other aspects and features of the invention are described with the intent to be illustrative, and not restrictive. This invention has been described herein with detail in order to comply with the patent statutes and to provide those skilled in the art with information needed to apply the novel principles and to construct and use such specialized components as are required. Further, those skilled in the art will recognize the components described and illustrated and will further appreciate other suitable components to perform the above-described features of the invention. It is to be understood, however, that the invention can be carried out by specifically different constructions, and that various modifications, both as to the construction and operating procedures, can be accomplished without departing from the scope of the invention. Further, in the appended claims, the transitional terms comprising and including are used in the open-ended sense in that elements in addition to those enumerated may also be present. Other examples will be apparent to those of skill in the art upon reviewing this document.

Claims

1. An apparatus for connecting a hand truck to a receiver mounted to a vehicle, the apparatus comprising: a lever arm coupled to a hand truck and having a free end portion extending outward from the hand truck; the free end portion is adapted to be received within a receiver; a glide that is separable from the lever arm and receiver, wherein the glide is adapted to be received within the

receiver; and wherein one of the glide and the lever arm includes a claw and wherein an other of the glide and the lever arm has a pivot member adapted to couple with the claw.
2. The apparatus as recited in claim 1, further including a base engaged to the hand truck wherein the lever arm engages the base, and further including an end portion of the lever arm that is slidingly extendable from the base.
3. The apparatus as recited in claim 2, further including a lever lock coupled to the base and having a first locked position that engages the lever arm and having a second retracted position that disengages the lever arm from the lock.
4. The apparatus as recited in claim 1, further including wear plates attached to sides of the glide.
5. The apparatus as recited in claim 1, further including wear plates attached to sides of the lever arm.
6. The apparatus as recited in claim 1, wherein the glide further includes an actuatable latch having an end portion extending outward from a side of the glide.
7. An apparatus for connecting a hand truck to a receiver of a vehicle, the apparatus comprising: a lever arm coupled to a base of a hand truck, the lever arm having a free end portion extending outward from the hand truck; a receiver adapted for coupling to a vehicle; a glide slidingly engaged within the receiver; a drive mechanism coupled to the glide to actuate the glide within the receiver between a loading position and a retracted position; and wherein one of the glide and the lever arm includes a claw and wherein an other of the glide and the lever arm has a pivot member adapted to couple with the claw.
8. The apparatus as recited in claim 7, further including at least two spaced apart winches coupled to at least one of the hand truck and the vehicle.
9. The apparatus as recited in claim 8, further including a controller electrically linked with the winches and drive mechanism, wherein the controller provides a signal to control coordinated activation of the winches and drive mechanism.
10. The apparatus as recited in claim 9, further including sensors positioned on the hand truck and the receiver.
11. The apparatus as recited in claim 10, wherein the sensors provide an output that is capable of being used to correlate a three-dimensional orientation of the hand truck and the receiver relative to gravity.
12. The apparatus as recited in claim 11, wherein the output from the sensors is transmitted to the controller.
13. The apparatus as recited in claim 7, wherein the drive mechanism includes a linear actuator.
14. The apparatus as recited in claim 13, wherein the linear actuator includes a lead screw, lead nut, and stepper motor.
15. The apparatus as recited in claim 7, wherein the glide includes roller bearings coupled to the glide.
16. An apparatus for connecting a hand truck to a receiver of a vehicle, the apparatus comprising: a lever arm coupled to a base of a hand truck, the lever arm having a free end portion extending outward from the hand truck; a receiver adapted for coupling to a vehicle; a glide slidingly engaged within the receiver; a drive mechanism coupled to the glide to actuate the glide within the receiver between a loading position and a retracted position; wherein one of the glide and the lever arm includes a claw and wherein an other of the glide and the lever arm has a pivot member adapted to couple with the claw; at least two spaced apart winches coupled to at least one of the hand truck and the vehicle; and a controller electrically linked with the winches and drive mechanism, wherein the controller provides a signal to control coordinated activation of the winches and the drive mechanism.
17. The apparatus as recited in claim 16, further including sensors positioned on the hand truck and the receiver.
18. The apparatus as recited in claim 17, wherein the sensors provide an output that is capable of

being used to correlate a three-dimensional orientation of the hand truck and the receiver relative to gravity.

19. The apparatus as recited in claim 18, wherein the output from the sensors is transmitted to the controller.

20. The apparatus as recited in claim 16, wherein the drive mechanism includes a linear actuator.